

XIAOCHUN NIU

Email: xiaochun.niu@duke.edu ◇ Website: xiaochunniu.github.io ◇ Tel: (+1) 773-961-5347

ACADEMIC APPOINTMENTS

Duke University, Durham, NC

The Fuqua School of Business

Postdoctoral Researcher in Decision Sciences

Advisor: Prof. Jiaming Xu

July 2024–Present

EDUCATION

Northwestern University, Evanston, IL

Department of Industrial Engineering and Management Sciences

Ph.D. in Operations Research

Advisors: Prof. Ermin Wei and Prof. Julia Gaudio

Sept. 2019–June 2024

Nanjing University, Nanjing, China

Department of Mathematics

B.S. in Mathematics

National Elite Program (Honors Program)

Sept. 2015–June 2019

RESEARCH INTERESTS

Operations research, applied probability, high-dimensional statistics, optimization, machine learning, artificial intelligence, data analytics, and network science.

PUBLICATIONS AND PREPRINTS

Journal Articles

- J1. “[High-Dimensional Procrustes Matching via Tree Counts](#).” Xiaochun Niu, Tselil Schramm, and Jiaming Xu. Preprint, 2026.
- J2. “[Optimality of Random Regular Graphs for Sparse Network Designs](#).” Weijia Li, Xiaochun Niu, Yehua Wei, and Jiaming Xu. *Major revision at Operations Research*, 2026.
 - Twenty-Seventh ACM Conference on Economics and Computation (EC 2026)
 - Oral presentation at INFORMS MSOM Annual Conference 2026
- J3. “[Learning with Shared Representations: Statistical Rates and Efficient Algorithms](#).” Xiaochun Niu, Lili Su, Jiaming Xu, and Pengkun Yang. *Minor revision at Journal of Machine Learning Research*, 2026.
 - Oral presentation at FL@FM–NeurIPS 2024
 - Oral presentation at INFORMS Workshop on Data Science 2026
- J4. “[Exact Label Recovery in Euclidean Random Graphs](#).” Julia Gaudio, Charlie Guan, Xiaochun Niu, and Ermin Wei. *Annals of Applied Probability*, 2026.
- J5. “[DISH: A Distributed Hybrid Optimization Method Leveraging System Heterogeneity](#).” Xiaochun Niu and Ermin Wei. *IEEE Transactions on Signal Processing*, 2024.
 - Extended abstract at Allerton Conference 2022

- J6. “FedHybrid: A Hybrid Federated Optimization Method for Heterogeneous Clients.” Xiaochun Niu and Ermin Wei. *IEEE Transactions on Signal Processing*, 2023.

Conference Proceedings

- C1. “Optimality of Random Regular Graphs for Sparse Network Designs.” Weijia Li, Xiaochun Niu, Yehua Wei, and Jiaming Xu. *ACM Conference on Economics and Computation (EC)*, 2026.
- C2. “Incentive Analysis for Agent Participation in Federated Learning.” Lihui Yi, Xiaochun Niu, and Ermin Wei. *IEEE Conference on Decision and Control (CDC)*, 2025.
- C3. “Exact Community Recovery in the Geometric Stochastic Block Model.” Julia Gaudio, Xiaochun Niu, and Ermin Wei. *ACM-SIAM Symposium on Discrete Algorithms (SODA)*, 2024.
- Nemhauser Prize for Best Student Paper, 2024
- C4. “Understanding Generalization of Federated Learning via Stability: Heterogeneity Matters.” Zhenyu Sun, Xiaochun Niu, and Ermin Wei. *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
- C5. “DISH: A Distributed Hybrid Primal-Dual Optimization Framework to Utilize System Heterogeneity.” Xiaochun Niu and Ermin Wei. *IEEE Conference on Decision and Control (CDC)*, 2022.

SELECTED HONORS AND AWARDS

• Rising Stars in Computational and Data Sciences, UT Austin	2024
• Nemhauser Prize for Best Student Paper, Northwestern University	2024
• SIAM Student Travel Award, SODA	2024
• Terminal Year Fellowship, Northwestern University	2023
• Arthur P. Hurter Award for Academic Excellence, Northwestern University	2020
• Walter P. Murphy Graduate Fellowship, Northwestern University	2019
• Excellent Graduate Award, Nanjing University	2019
• UCLA CSST Research Program Scholarship, University of California, Los Angeles	2018
• National Scholarship (<i>Top 2%</i>), Nanjing University	2016

SELECTED TALKS AND PRESENTATIONS

Optimality of Random Regular Graphs for Sparse Network Designs

• INFORMS Annual Meeting, San Francisco, CA	Nov. 2026
• INFORMS MSOM Conference, Boston, MA	July 2026
• INFORMS Annual Meeting, Atlanta, GA	Oct. 2025

Learning with Shared Representations: Statistical Rates and Efficient Algorithms

• INFORMS, Workshop on Data Science, San Francisco, CA	Oct. 2026
• INFORMS Applied Probability Society Conference, Atlanta, GA	June 2025
• International Workshop on Federated Foundation Models (NeurIPS 2024), Vancouver	Dec. 2024
• INFORMS Annual Meeting, Seattle, WA	Oct. 2024
• Annual Allerton Conference on Communication, Control, and Computing (Allerton), UIUC	Sept. 2024

Exact Community Recovery in the Geometric Stochastic Block Model

- IEMS-SIAM Student Seminar, Northwestern University *May 2024*
- Rising Stars in Computational and Data Sciences, UT Austin *April 2024*
- Institute for Data, Econometrics, Algorithms, and Learning (IDEAL) Special Program on Network and Inference, Toyota Technological Institute at Chicago (TTIC) *Jan. 2024*
- ACM-SIAM Symposium on Discrete Algorithms (SODA), Alexandria, VA *Jan. 2024*

DISH: A Distributed Hybrid Optimization Method Leveraging System Heterogeneity

- INFORMS Annual Meeting, Phoenix, AZ *Oct. 2023*
- INFORMS Annual Meeting, Indianapolis, IN *Oct. 2022*
- INFORMS Optimization Society Conference, Greenville, SC *Mar. 2022*

TEACHING EXPERIENCE

Duke University

- EGRMGMT 580, Decision Models, Instructor *Spring 2026*

Northwestern University

- IEMS 435, Stochastic Simulation, Grader *Spring 2022*
- IEMS 344, Whole Brain Leadership, Teaching Assistant *Fall 2021*

INDUSTRY EXPERIENCE

Meta Platforms

Menlo Park, CA

Research Scientist Intern, Social Impact Machine Learning Team

June 2022–Sept. 2022

- Improved Facebook's donation-specific newsfeed ranking algorithm by implementing new features and optimizing neural network architecture.

The Home Depot

Atlanta, GA (remote)

Collaborative Data Project

Sept. 2022–Mar. 2023

- Evaluated multi-armed bandit methods on dynamic ad placement datasets with frequent arm adjustments.

PROFESSIONAL SERVICE AND MENTORING

- Reviewer:
 - Journals: Operations Research, Annals of Statistics, Annals of Applied Probability, Journal of Machine Learning Research, Journal of Computational and Graphical Statistics, IEEE Transactions on Information Theory, IEEE Transactions on Automatic Control, IEEE Transactions on Signal Processing, IEEE Transactions on Signal and Information Processing over Networks
 - Conferences: SODA 2026, IEEE Conference on Decision and Control, Midwest Machine Learning Symposium, American Control Conference
- Session Organizer: IEEE CDC 2022 (Distributed Optimization and Learning for Networked Systems)
- Mentor: NCSSM Mentorship Program 2026, IDEAL Get Ready for Research Workshop 2023, INFORMS Mentorship Program 2023, Northwestern English Language Program 2022